

Mostafa Zaki

Big Data Engineer

mostafa.zaki@email.com | +20 100 123 4567 |  github.com/mostafa-zaki |  linkedin.com/in/mostafa-zaki | mostafazaki.com

Education

Medi-caps Institute of Technology & Management

CGPA: 3.86

B.E. Electronics & Communication

May 2015, Indore, Madhya Pradesh, India

Experience

Big Data Engineer

Full-time

Tata Consultancy Services

August 2015 - Present, Cairo, Egypt

- Led the Data Lake program, creating a secure data hub on Hadoop for data governance, lineage, auditing, and monitoring.
- Developed a proactive Health Check utility for Data Lake, monitoring Hadoop components and automating HTML email reports for daily checks and post-upgrades.
- Engineered data pipelines to ingest and process various data formats in the Hadoop ecosystem, applying appropriate filtering techniques.
- Developed a robust data pipeline utility for RDBMS to Hive external table ingestion using Sqoop, incorporating data quality checks (e.g., row count validation).
- Automated cluster health, usage, and capacity reporting through Unix shell scripting, enhancing operational oversight.
- Optimized Hive queries significantly to boost performance and minimize Hadoop resource utilization across projects.
- Configured Flume agents for efficient data processing and transfer into the Hadoop ecosystem.
- Conducted benchmark testing on Hive and Impala queries to ensure optimal performance and scalability.
- Contributed to Hadoop cluster setup, including edge node configuration and High Availability (HA) implementation for critical services (Hive Server2, Impala, HDFS).
- Executed data filtering from large datasets using technologies such as Pig and Regex Serde, addressing specific business requirements.
- Performed Dataiku benchmark testing on Impala and Hive, comparing performance against Greenplum databases.
- Managed data migration from Greenplum database to Hadoop via Sqoop pipelines, processing and storing data in Hive for performance evaluations.
- Resolved Hadoop ecosystem-related issues, ensuring stability for the WM Hadoop ecosystem.
- Rescheduled jobs from autosys to TWS hosting for improved performance and reliability.
- **Project: ABSA- Reconciliations, UBC and WMDATALAKE COE (Clients: Barclays, Union Bank of California, Morgan Stanley)**
- Optimized SQL queries for data not slated for migration from RDBMS, improving legacy system efficiency.
- Designed and implemented Hive queries and logic for seamless data migration from RDBMS to the Hadoop ecosystem.
- Authored Hive queries for in-depth data analysis, aligning with diverse business requirements.
- **Project: Intellimatch Migration**
- Developed and implemented Recs based on extracted business requirements, driving enhancements in reconciliation processes.
- Successfully migrated static reconciliation setups from Intellimatch version 8.1 to 9.1, ensuring system compatibility and upgrades.
- Executed backend tasks involving SQL query writing to provide essential data for new reconciliation configurations.

Initial Learning Program Trainee

Full-time

Tata Consultancy Services

June 2015 - August 2015, Indore, India

- Participated in a comprehensive learning program designed for new hires to align with organizational working standards.
- Received grooming in various technical and ethical aspects, foundational for professional development within the organization.

Skills

Programming Languages: Python - Shell Scripting - SQL - Scala

Big Data Frameworks/Technologies: Apache Hadoop - Hive - Pig - Sqoop - Flume - Oozie - Impala - HDFS - MapReduce - Spark - Unix

Tools & Platforms: Dataiku - Unravel - Cloudera - Putty - HUE - Cloudera Manager - Eclipse - Resource Manager - Greenplum